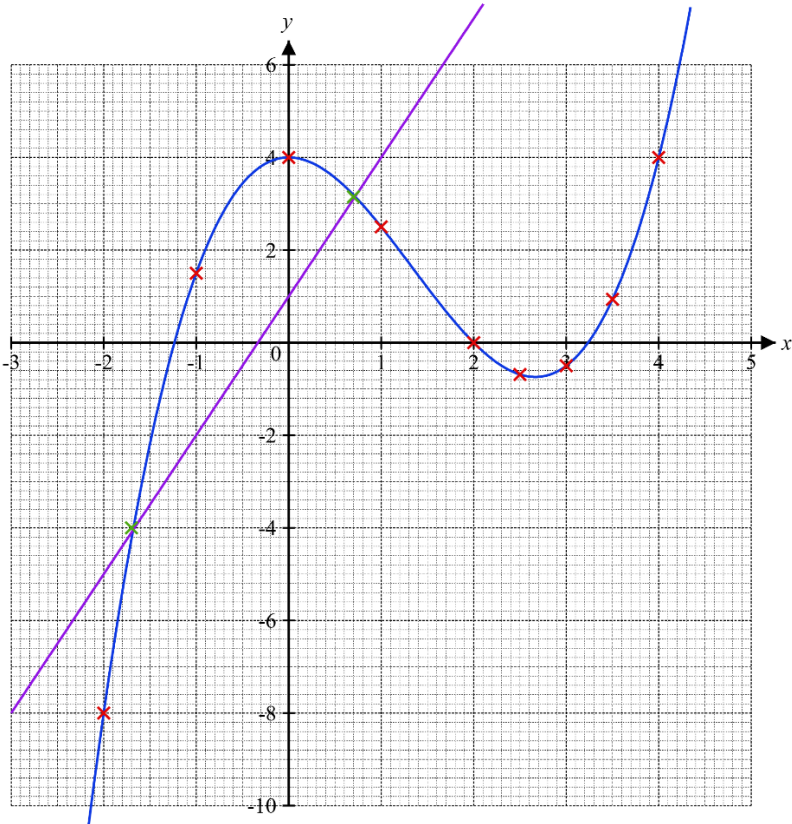


Bedok South Secondary School
Mathematics Department
Secondary 4 Exp Prelim P2 2026
Mark Scheme

Question		Solution
1	a	$3(2x - 5) - 4(x + 7)$ $= 6x - 15 - 4x - 28$ $= 2x - 43$
	b	$42a^2b^3 + 56ab^2$ $= 14ab^2(3ab + 4)$
	ci	$t = kd^2$ $9.375 \times 10^2 = k(2.5 \times 10^3)^2$ $k = \frac{9.375 \times 10^2}{(2.5 \times 10^3)^2}$ $= 1.5 \times 10^{-4}$ $= 1.5 \times 10^{-4}$ $t = 1.5 \times 10^{-4} \times d^2$
	cii	$2 \times 10^3 = t = 1.5 \times 10^{-4} \times d^2$ $d^2 = \frac{2 \times 10^3}{1.5 \times 10^{-4}}$ $d = \pm \sqrt{\frac{2 \times 10^3}{1.5 \times 10^{-4}}}$ $= \pm 3651.4837...$ $= \pm 3.65 \times 10^3$

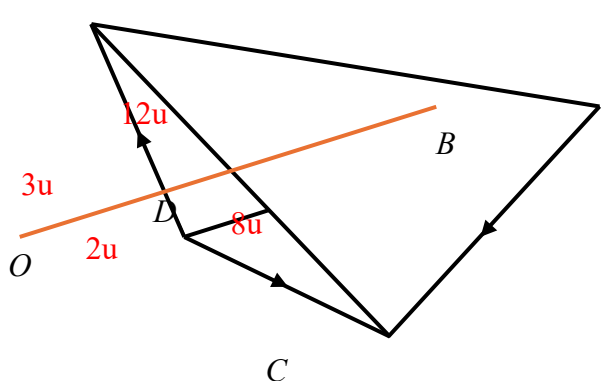
2	a	Since number of data is even, $\text{Median} = \frac{y+11}{2} = 12$. $y = 13$ Since mode is 18, $z = 18$ $\text{Mean} = 12 = \frac{25+18+18+16+13+11+9+x+5+2}{10}$ $120 = 117 + x$ $x = 3$
	bi	60% success rate means $\frac{60}{100} \times 15 = 9$ successful free throws $P(\text{selected player has at least 9 makes}) = \frac{10}{16} = \frac{5}{8} = 0.625 = 62.5\%$
	bii	If 13 were selected, 3 were not selected. $m = 7 + 1 = 8$
3	a	By drawing another N at B, notice alt angles Bearing of dock from B $= 232 - 180$ (alt $\angle$, parallel Norths) $= 052^\circ$
	b	$\angle ADB = 318 - 232 = 86$ By cosine rule, $AB^2 = 22^2 + 30^2 - 2(22)(30)\cos 86^\circ$ $AB = 35.9433...$ ≈ 35.9
	c	$\text{Area} = \frac{1}{2}(22)(30)\sin 86^\circ$ $= 329.1961...$ Since $329 > 300$, I agree with the captain to seek assistance.

4	a	17, 21
	bi	$S_n = 5 + (n-1) \times 4$ $= 5 + 4n - 4$ $= 4n + 1$
	bii	$U_n = (2n)^2 = 4n^2$
	c	$U_{81} = 4(81)^2 = 26244$
	d	Let $S_n = 2026$ $4n + 1 = 2026$ $4n = 2025$ $n = \frac{2025}{4} = 506.25$ Since $n = 506.25$ is not a positive integer, no figure in the sequence has 2026 shaded squares.
5	a	$1^3 : 2^3$ $= 1 : 8$
	b	The spheres show that center Q is using 8 times the water that center P is while the actual usage is only 2 times.
	c	$\left(\frac{3}{r}\right)^3 = \frac{1}{2}$ $\frac{3}{r} = \sqrt[3]{\frac{1}{2}}$ $r = 3.77976... \approx 3.78$

6	a	$-0.7, 0.9$
	b	
	c	Graphical: $-0.8 < k < 4$ (Theoretical: $-0.7407... < k < 4$)
	d	$x^3 - 4x^2 - 6x + 6 = 0$ $\frac{x^3}{2} - 2x^2 - 3x + 3 = 0$ $\frac{x^3}{2} - 2x^2 + 3 = 3x$ $\frac{x^3}{2} - 2x^2 + 3 + 1 = 3x + 1$ $\frac{x^3}{2} - 2x^2 + 4 = 3x + 1$ Plot $y = 3x + 1$ From graph, $x = -1.7$ or 0.7

7	a	$\angle BDE = 180 - 124$ (opp. $\angle$ in cyclic quad) $= 56$ $\angle BOE = 56 \times 2$ ($\angle$ at center $= 2 \times \angle$ at circumference) $= 112^\circ$
	b	$\angle FBO = \angle FEO = 90^\circ$ (tangent $\perp$ radius) $\angle BFE = 360 - 90 - 90 - 112$ ($\angle$ sum in quad) $= 68^\circ$
	c	Draw line BE to form isos triangle OBE $\angle OBE = \angle OEB$ $= \frac{180 - 112}{2}$ (base $\angle$ in isos. Δ) $= 34$ Consider triangle BDE , $\angle GBO + \angle GEO$ $= 180 - 56 - 34 - 34 - 15$ ($\angle$ sum in Δ) $= 41^\circ$

8	a	$\begin{pmatrix} 500 & 800 & 200 \\ 300 & 1200 & 400 \end{pmatrix}$
	b	$\begin{pmatrix} 500 & 800 & 200 \\ 300 & 1200 & 400 \end{pmatrix} \begin{pmatrix} 40 \\ 15 \\ 25 \end{pmatrix}$ $= \begin{pmatrix} 500(40) + 800(15) + 200(25) \\ 300(40) + 1200(15) + 400(25) \end{pmatrix}$ $= \begin{pmatrix} 37000 \\ 40000 \end{pmatrix}$
	c	They represent the total value (OR prices) of each portfolio, A and B in October.
	d	$\begin{pmatrix} 1.5 \\ 0.4 \\ 1.0 \end{pmatrix}$
	e	$\begin{pmatrix} 500 & 800 & 200 \\ 300 & 1200 & 400 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.4 \\ 1.0 \end{pmatrix}$ $= \begin{pmatrix} 500(1.5) + 800(0.4) + 200(1.0) \\ 300(1.5) + 1200(0.4) + 400(1.0) \end{pmatrix}$ $= \begin{pmatrix} 1270 \\ 1330 \end{pmatrix}$ Portfolio B is classified as “High Risk” while Portfolio A is not.
	f	$k \begin{pmatrix} 1270 \\ 1330 \end{pmatrix} = \begin{pmatrix} 1270k \\ 1330k \end{pmatrix}$ $1270k < 1000 \text{ and } 1330k < 1000$ $k < \frac{1000}{1270} \text{ and } k < \frac{1000}{1330}$ $k < 0.78740... \text{ and } k < 0.75187...$ $k < 0.75187$ Largest $k = 0.75$

9	a	$\overrightarrow{AC} = \mathbf{q} - \mathbf{p}$ $\overrightarrow{AD} = \frac{3}{5}(\mathbf{q} - \mathbf{p})$ $\overrightarrow{OD} = \mathbf{p} + \frac{3}{5}(\mathbf{q} - \mathbf{p})$ $= \mathbf{p} + \frac{3}{5}\mathbf{q} - \frac{3}{5}\mathbf{p}$ $= \frac{2}{5}\mathbf{p} + \frac{3}{5}\mathbf{q} = \frac{1}{5}(2\mathbf{p} + 3\mathbf{q})$
	b	$\overrightarrow{OB} = \mathbf{q} + 2\mathbf{p} + 2\mathbf{q}$ $= 2\mathbf{p} + 3\mathbf{q}$ $= 5\left(\frac{1}{5}(2\mathbf{p} + 3\mathbf{q})\right)$ $= 5\overrightarrow{OD}$ Since $\overrightarrow{OB}$ and $\overrightarrow{OD}$ are scalar multiples, OB and OD are parallel. Since OB and OD share a common point O, O, D and B lie on a straight line.
	c	 Since $AD : DC = 3 : 2$, $\frac{\text{Area of } \triangle OAD}{\text{Area of } \triangle OCD} = \frac{3}{2}$ Since $OD : DB = 1 : 4$ $\frac{\text{Area of } \triangle OAD}{\text{Area of } \triangle ABD} = \frac{\text{Area of } \triangle OCD}{\text{Area of } \triangle BCD} = \frac{1}{4}$ So, $\frac{\text{Area of } \triangle OAD}{\text{Area of } \triangle ABD} = \frac{3}{12}$, $\frac{\text{Area of } \triangle OCD}{\text{Area of } \triangle BCD} = \frac{2}{8}$ $\frac{\text{Area of } \triangle OAD}{\text{Area of } \triangle ABC} = \frac{\text{Area of } \triangle OAD}{\text{Sum of Area of all } \triangle} = \frac{3}{3+2+8+12} = \frac{3}{25}$

10	a	Volume of ceramic used for 1 mug $= \pi(4.5)^2 \times 10 - \pi(4)^2 \times 10$ $= \frac{85}{2} \pi = 133.517...$ Mass of ceramic used for 1 mug $= \frac{85}{2} \pi \times 2.9$ $= \frac{493}{4} \pi = 387.201... \approx 387$
	b	Diameter of each mug = $2 \times 4.5 = 9$ cm Since thickness of honeycomb wrap is 0.4 cm, Min length of packaging box required $= 4 \times (9 + 2 \times 0.4)$ $= 39.2$ cm Min breadth of packaging box required $= 2 \times (9 + 2 \times 0.4)$ $= 19.6$ cm Min height of packaging box required $= 10$ cm The company should choose the Medium Carton.

	c	Cost of ceramic $= \frac{493}{4} \pi \times 8 \div 100 \times 0.37$ $= \$11.4611\dots$ Cost of packaging box = \$6.10 Exterior circumference of mug = $9\pi = 28.274\dots$ which is more than 15 cm but less than 30 cm and height of mug is 10 cm which is less than 12 cm, 2 pieces of honeycomb wrap is required per mug Cost of honeycomb wrap = $8 \times 2 \times 0.45 = \\7.20 Total mass of package = Mass of ceramic + Mass of packaging box + Mass of Honeycomb Wrap $= 986\pi + 260 + 8 \times 2 \times 5$ $= 3437.610\dots$ g Air Postage to Canada $= 20 + 9 + 9 = \\$38$ Total cost incurred = Cost of Ceramic + Cost of Packaging Box + Cost of Honeycomb Wrap + Air Postage $= 11.4611\dots + 6.10 + 7.20 + 38$ $= \\$62.7611\dots$ Price to charge Jerry $= 1.25 \times 62.7611\dots$ $= 78.4514$ $\approx \\$78.45$ Company M should charge Jerry \$78.45
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